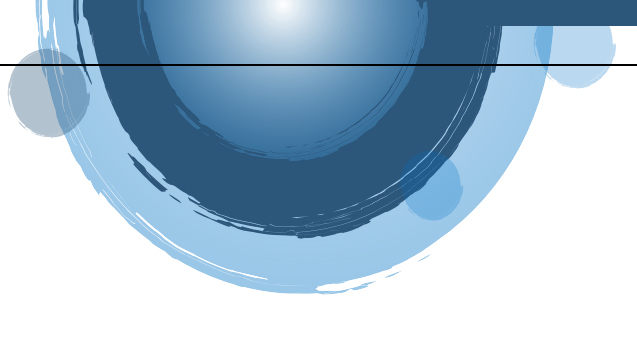




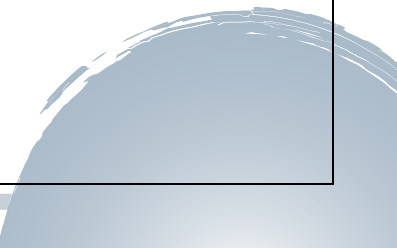
Cyber Security Internship Report

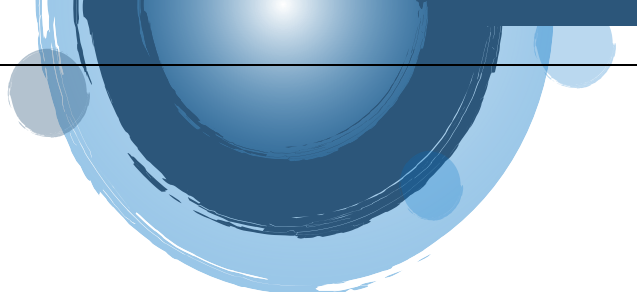
Cyber Security Internship – Task 16

Security Breach Simulation – Linux Log Based Incident Handling

Submitted by: Hetal Antala

Date : 14 / 02 / 2026



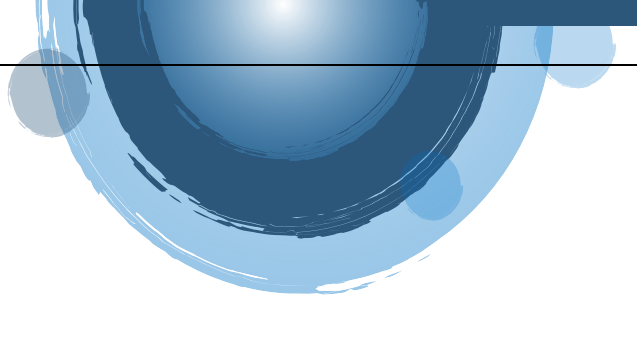


1. Introduction

Cybersecurity incidents such as unauthorized access attempts, brute-force attacks, and credential misuse are common threats faced by modern systems. Organizations must be prepared to detect, analyze, contain, and recover from such attacks quickly to minimize potential damage.

This project demonstrates a practical simulation of a security breach and the implementation of a structured incident response process using Linux authentication logs. The simulation involved generating repeated failed login attempts to mimic a brute-force attack and then applying standard incident handling techniques to investigate and secure the system.

The objective was to gain hands-on experience with real-world incident response procedures and security monitoring.



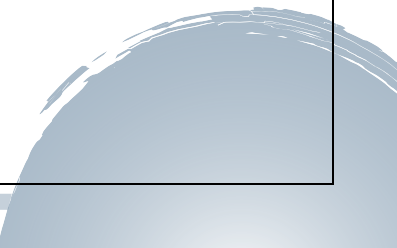
2. Objectives

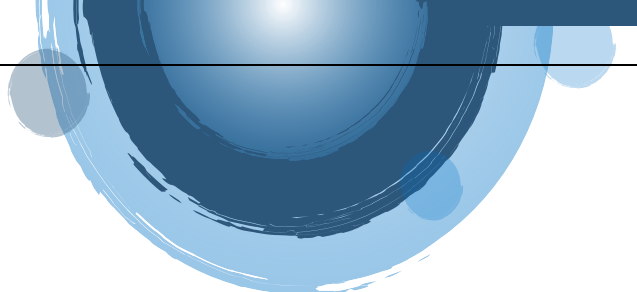
The primary objectives of this task were:

- Simulate a security incident (failed login attempts)
 - Monitor and analyze system logs
 - Detect suspicious behavior
 - Classify the attack type and severity
 - Contain the threat
 - Remove the root cause
 - Restore system integrity
 - Document the entire process
 - Recommend preventive measures
-

3. Tools Used

The following tools were used during this simulation:





Tool

Purpose

Linux Terminal

Command execution

/var/log/auth.log

Authentication monitoring

lastb command

Failed login tracking

passwd

Account management

ufw firewall

Blocking malicious IPs

GitHub

Documentation &
submission

4. System Environment

Component

Details

OS

Linux (Kali/Ubuntu/WSL)

User Account

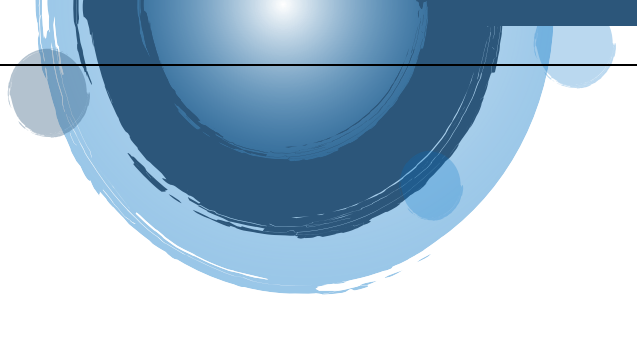
kali

Log Type

Authentication logs

Shell

Bash

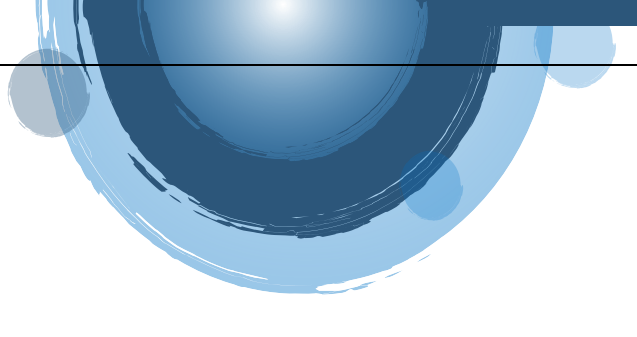


5. Incident Scenario Description

To simulate a real-world security breach, multiple failed login attempts were intentionally generated using incorrect credentials. This mimicked a brute-force or password guessing attack where an attacker tries multiple passwords to gain unauthorized access.

The system's authentication logs were monitored to detect unusual login patterns. Repeated failures within a short time period triggered suspicion of malicious activity.

This scenario reflects common real-world attacks on SSH, sudo, or user login services.

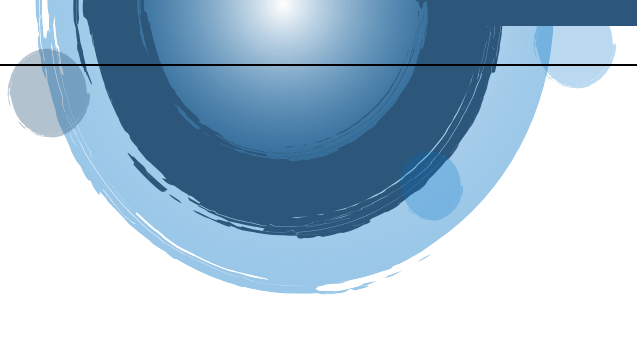


6. Incident Response Methodology

The standard Incident Response lifecycle was followed:

- 1.Detection
- 2.Identification
- 3.Classification
- 4.Containment
- 5.Eradication
- 6.Recovery
- 7.Documentation
- 8.Prevention

Each phase is explained below.



7. Detection Phase

During system monitoring, abnormal authentication behavior was observed. Multiple login failures were recorded for the same user account within a short time span.

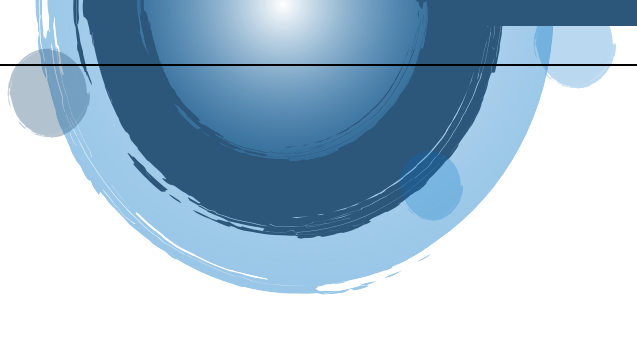
Commands used:

```
sudo cat /var/log/auth.log | grep "Failed"
```

Findings:

- More than 10 failed attempts
- Same account targeted repeatedly
- Short intervals between attempts
- Behavior consistent with brute-force attack

This confirmed suspicious activity.



8. Identification and Analysis

Log analysis revealed:

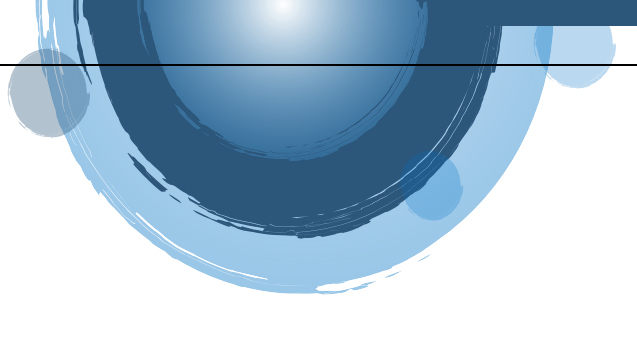
- Timestamp of attempts
- Targeted username
- Authentication failures
- Repeated attempts from the same source

Example log evidence:

Failed password for kali from 192.168.1.15 port 22 ssh2

Analysis indicated:

- Automated attack pattern
 - Credential guessing
 - Unauthorized access attempt
-



9. Incident Classification

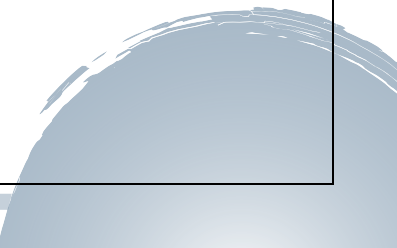
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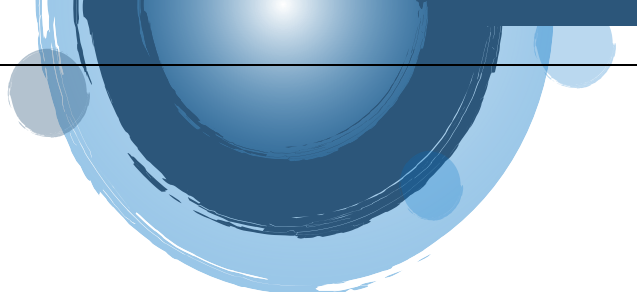
Parameter	Value
Incident Type	Brute Force Attack
Target	User account
Impact Level	Medium
Risk Level	High if successful
Status	Contained

The incident was classified as medium severity because no successful breach occurred, but the risk of compromise was high.

10. Containment Phase

The primary goal during containment is to stop the attacker immediately.





Actions taken:

1. Lock compromised account

sudo passwd -l kali

This disabled login access temporarily.

2. Block suspicious IP

sudo ufw deny from <suspicious_ip>

This prevented further connection attempts.

3. Monitor logs continuously

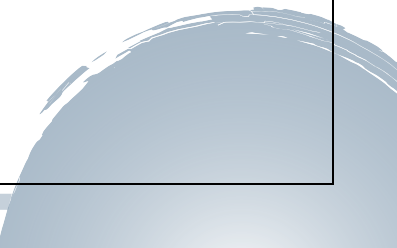
Ensured no new attempts were occurring.

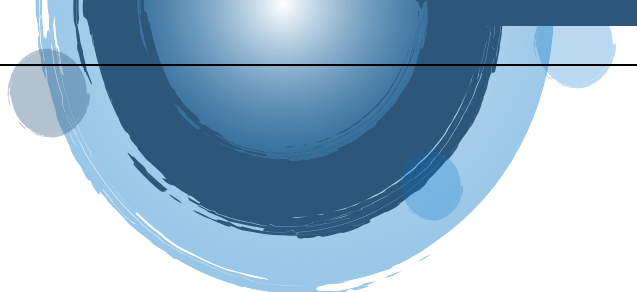
Containment prevented further damage.

11. Eradication Phase

After containment, the threat was removed.

Steps performed:

- Password changed
 - Verified no unauthorized users created
 - Cleared suspicious sessions.
- 

- 
- Verified services integrity

Commands:

sudo passwd kali

sudo passwd -u kali

The system was cleaned and secured.

12. Recovery Phase

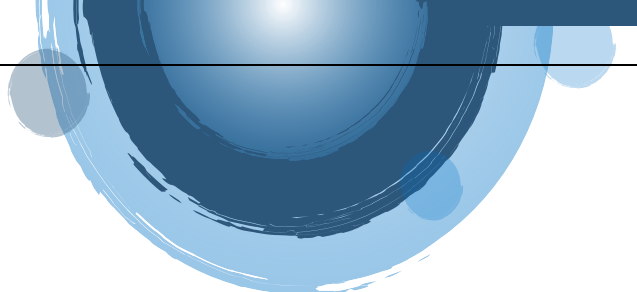
System functionality was restored.

Activities:

- Account unlocked
- Normal login verified
- Services restarted
- Continuous monitoring enabled

No further suspicious activity was observed after remediation.

The system returned to a safe operational state.



13. Root Cause Analysis

Root cause investigation identified:

Primary cause:

Weak authentication protection allowing repeated login attempts.

Contributing factors:

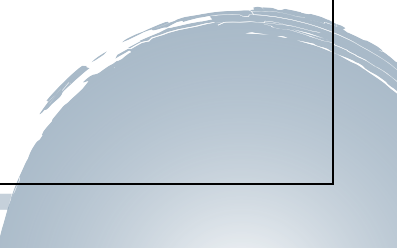
- No rate limiting
- No account lockout policy
- Lack of monitoring initially

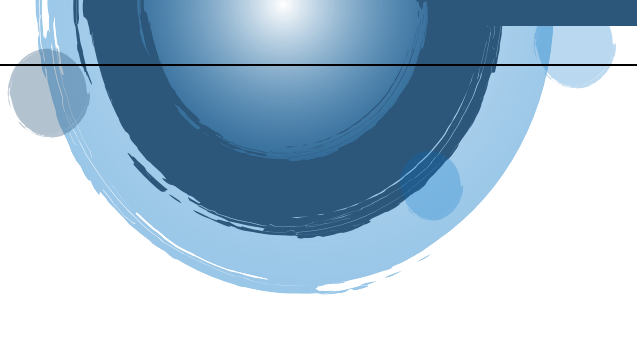
Risk:

Attackers could eventually guess password if unchecked.

14. Evidence Collected

Evidence supporting the investigation:



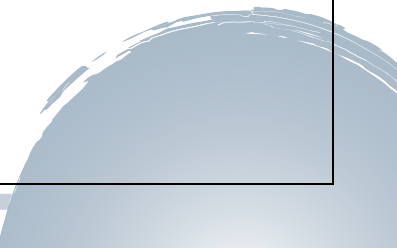
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- Authentication logs
 - Failed login records
 - Terminal outputs
 - Screenshots
 - Firewall rules

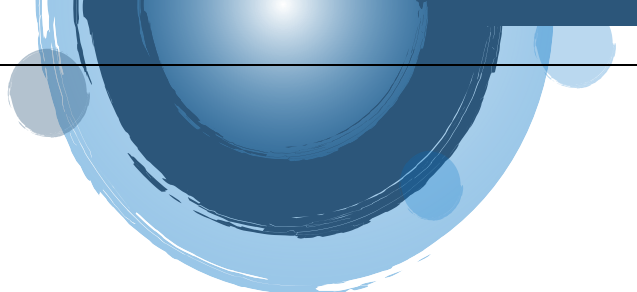
All evidence stored in the screenshots directory.

15. Preventive Measures and Recommendations

To prevent similar incidents, the following measures are recommended:

Security Hardening

- Strong passwords
 - Multi-factor authentication
 - Account lockout policies
 - Disable unused accounts
- 



Monitoring

- Regular log analysis
- Real-time alerts
- Intrusion detection systems

Tools

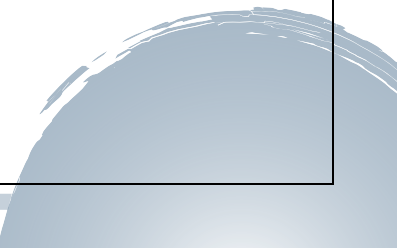
- Fail2ban
- Firewall configuration
- Log monitoring automation

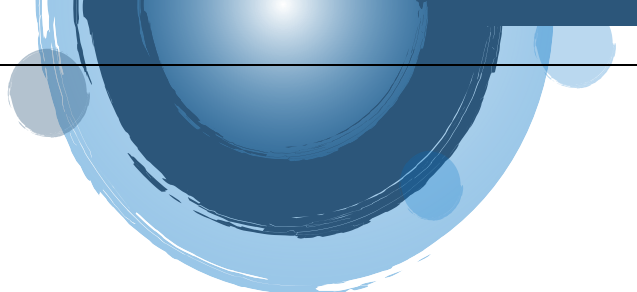
Best Practices

- Regular security audits
- Patch management
- User awareness training

16. Lessons Learned

Through this simulation, the following skills were developed:

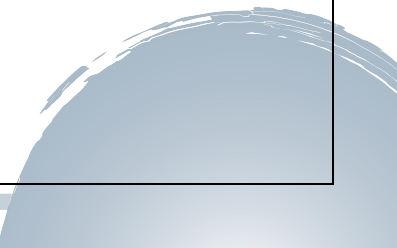


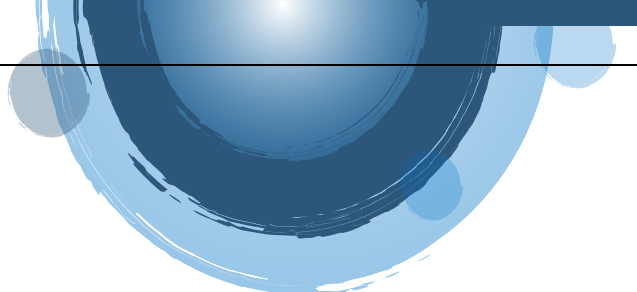
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- Log analysis techniques
 - Threat detection
 - Incident classification
 - Containment strategies
 - Root cause analysis
 - System recovery procedures
 - Professional documentation

This task provided practical exposure to real-world SOC operations.

17. Conclusion

The incident response simulation successfully demonstrated the complete lifecycle of handling a security breach. The brute-force attack was detected early, contained promptly, and mitigated effectively without system compromise.





The exercise improved understanding of defensive security practices and incident management workflows.

Proper monitoring and preventive measures can significantly reduce the risk of successful attacks.

